Evaluation Report of Elite_FS_Day_21_EigenVectors-KMIT

Average Score: 4/10 | Attempted: 3 of 3 questions

Question 1

THEORY UNDERSTAND Best Score: 5/10

What is Data Drift? Explain different types of data drift.

Desired Answer

Data drifting (or data drift) refers to a change in the statistical properties of input data over time compared to the data used to train a machine learning model. When the underlying data distribution shifts, the model's predictions may become less accurate because it was trained under different conditions. Formally, if a model was trained on data drawn from distribution Ptrain(X), and over time the incoming data follows a different distribution Plive(X), then: Ptrain(X)□= Plive(X) This difference is called data drift. Types of Data Drift 1. Covariate Drift (Feature Drift): The distribution of input features changes, but the relationship between input and output remains the same. 2. Concept Drift: The relationship between input features and the target variable changes over time. 3. Label Drift (Prior Probability Shift): The distribution of the target variable changes

Student's Answer

Data Drift is the phenomenon where the data has been drifted from the original training data. for example: the number of users of chatgpt in 2022 is less as compared to its users in 2026. here we can clearly see a huge change in the numbers, similarly if the data changes overs the years it leads to the model performing worse and worse over the years. it can be different types like mean shift, variance shift, etc..

Evaluations

Feedback 1

★★½ ☆☆ 5/10

The student is addressing the same concept of data drift (a change in data over time leading to model degradation), so the response is relevant. They correctly identify that data changing over years can harm model performance and give a simple example. However, they do not formalize the definition in terms of shifting distributions (Ptrain versus Plive) and fail to name or explain the three types of drift from the Actual Answer (covariate drift, concept drift,

label/prior probability shift). Instead they mention generic "mean shift" and "variance shift," which do not align with the specified categories. This shows partial understanding of the core idea but misses key details and the required types, placing this at an Adequate Understanding level.

Feedback 2

★½ ☆☆☆ 3/10

The student is addressing the same topic of data drift, so the response is relevant. They correctly identify that data drift means the input data distribution changes over time and note its effect on model performance. However, their explanation is very high-level and they fail to explain the specific types of drift named in the Actual Answer (covariate drift, concept drift, and label drift). Instead, they vaguely refer to 'mean shift' and 'variance shift' without defining or relating them to the formal categories. This shows a fragmentary understanding: they grasp the basic idea of drift but miss most of the key distinctions and formal definitions required. Hence, the answer demonstrates weak understanding with significant gaps.

Question 2

THEORY REMEMBER Best Score: 1/10

What is KL Divergence?

Desired Answer

Kullback–Leibler (KL) Divergence is a measure from information theory that quantifies how one probability distribution differs from a reference probability distribution. It represents the amount of information lost when an approximate distribution Q is used to represent the true distribution P. For discrete probability distributions, KL divergence is defined as: $D_{KL}(P\|Q)=\sum P(x)\log(P(x)/Q(x))$ Conceptually, KL divergence measures the expected extra number of bits required to encode samples from distribution P when using a coding scheme optimized for Q instead of P. Thus, it quantifies the divergence in information content between two probability distributions.

Student's Answer

It is a metric used in encoders. It involves having to calculate integrals and logs in a very complicated manner.

Evaluations

Feedback 1

1/10

The student answer provides almost no correct information from the Actual Answer. It calls KL divergence merely a 'metric used in encoders' and vaguely mentions 'integrals and logs,' but makes no reference to comparing two probability distributions P and Q, the formula $D_{KL}(P||Q)=\sum P(x)\log(P(x)/Q(x))$, or the interpretation as extra bits of information. Nearly all key facts from the Actual Answer are missing or incorrect, so this is very poor recall.

Feedback 2

0/10

Fail Condition: Generic Copy-Paste / Minimal Effort. The student's answer does not recall any of the core facts from the Actual Answer: there is no mention of probability distributions P and Q, quantification of divergence, information theory context, the KL formula, or the interpretation in bits. Instead, it only vaguely refers to encoders and complicated calculations, which is irrelevant to the defining characteristics of KL Divergence.

Question 3

THEORY UNDERSTAND Best Score: 5/10

What is the significance of eigenvectors for an AI engineer?

Desired Answer

Eigenvectors are significant for an AI engineer because they identify the fundamental directions along which data or transformations behave most strongly. Mathematically, for a matrix A, an eigenvector v satisfies $Av=\lambda v$, meaning the transformation scales the vector without changing its direction. This helps reveal the intrinsic structure of data and models. In practical AI applications, eigenvectors are central to Principal Component Analysis (PCA), where they determine the directions of maximum variance for dimensionality reduction and feature extraction. They also play a crucial role in analyzing model stability and optimization, particularly through Hessian eigenvalues, which indicate whether a model is converging to sharp or flat minima. Additionally, eigenvectors are used in spectral clustering, graph-based learning, Markov processes, and representation learning, where they uncover hidden patterns, clusters, and steady-state behaviors. Overall, eigenvectors provide a mathematical foundation for understanding data geometry, learning dynamics, and structural properties of AI systems.

Student's Answer

Eigen Vectors are the vectors that satisfy the equation of Av = lambda*v. They give the natural direction of variations in the data, keeping the big ones and drop noise, which results in faster training and better features. They control how signals and gradients move through layers, so preventing vanishing gradient so guide better weight initializations and optimizers. They also capture meaningful patterns.

Evaluations

Feedback 1

★★½☆☆ 5/10

The student is addressing the same topic—why eigenvectors matter for AI engineers—and correctly states the mathematical definition Av = λv and their role in capturing natural directions of variation (as in PCA for feature extraction and noise reduction). However, they miss several key applications from the Actual Answer (e.g., Hessian eigenvalues for model stability, spectral clustering, graph-based learning, Markov processes, representation learning) and introduce an incorrect claim about eigenvectors preventing vanishing gradients through weight initialization. This shows a basic but incomplete and partially inaccurate grasp of the full significance of eigenvectors in AI systems.

Feedback 2

★★½☆☆ 5/10

The student is addressing the same concept of eigenvectors (Av = λv) and their role in revealing principal directions of data variation (as in PCA), so the response is relevant. They correctly state the definition and recognize that eigenvectors identify dominant variation directions and help in feature extraction. However, they omit other key applications mentioned in the Actual Answer—such as use in Hessian analysis for model stability and optimization, spectral clustering, graph-based learning, Markov processes, and representation learning—and introduce an unsupported claim about controlling gradient flow and preventing vanishing gradients. Overall, the answer shows a basic grasp of eigenvectors and PCA but misses or misunderstands several major points from the Actual Answer.